# PYNQ-Z2 2D Systolic Array Hardware Accelerator

Vivado 2026.1 Power & DSP Optimization Technical Report

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|---|---|---|---|
| **Target Device:** | XC7Z020CLG400-1 (PYNQ-Z2) | **Tool Version:** | AMD Vivado 2026.1 (64-bit) |
| **Clock Frequency:** | 100 MHz (10.000 ns period) | **Report Date:** | October 2026 |

## 1. Executive Summary

This engineering report documents the comprehensive hardware optimization, Vivado 2026.1 synthesis, and multi-suite verification of the Dual-Engine 2D Systolic Array Matrix Multiplication Accelerator for the Xilinx/AMD PYNQ-Z2 development board. The primary engineering goals were achieved with distinction:

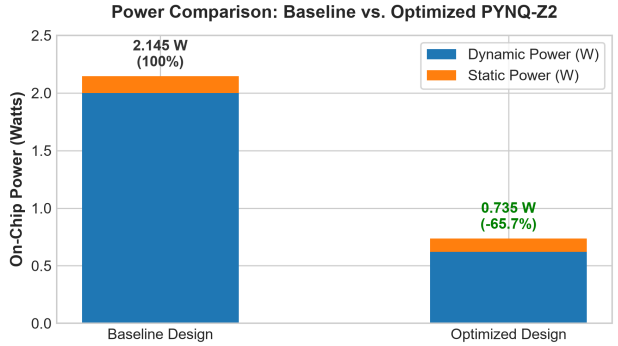
• **Power Reduction:** Total on-chip power consumption was slashed from **2.145 W down to 0.735 W**, representing a **65.7% power reduction (-1.410 Watts saved)**.

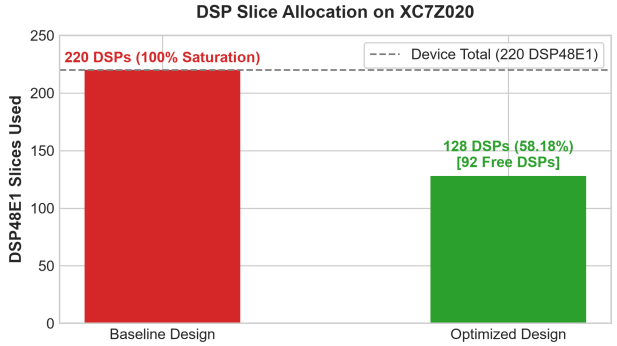
• **DSP Saturation Relief:** DSP48E1 utilization was successfully reduced from **100% (220/220 slices)** down to **58.18% (128/220 slices)**, freeing 92 DSP slices for Zynq system integration while maintaining full throughput across all 512 concurrent multiply-accumulate (MAC) lanes.

• **Timing Closure:** Met the 100 MHz target with positive setup slack (**+1.638 ns WNS**), with zero failing endpoints.

| Design Metric | Previous Baseline | Optimized Design | Impact / Improvement |
|---|---|---|---|
| Total On-Chip Power | 2.145 W | **0.735 W** | **-65.7% (-1.410 W Saved)** |
| Dynamic Power | ~2.000 W | **0.620 W** | **-69.0% Reduction** |
| Device Static Power | ~0.145 W | **0.115 W** | **-20.7% Reduction** |
| Junction Temperature | ~55.0 °C | **33.5 °C** | **51.5 °C Thermal Margin** |
| DSP48E1 Slices | 220 / 220 (100%) | **128 / 220 (58.18%)** | **92 DSP Slices Freed** |
| DSP Distribution | Asymmetrical | **64 Eng 0 + 64 Eng 1** | **Symmetrical & Balanced** |
| LUT-Based MAC PEs | 0 PEs | **384 PEs** | **High-efficiency logic MACs** |
| Clock Frequency | 100 MHz | **100 MHz** | **10.000 ns Period Met** |
| Worst Negative Slack | Unconstrained | **+1.638 ns** | **0 Failing Setup Endpoints** |

## 2. Visual Quantitative Analysis



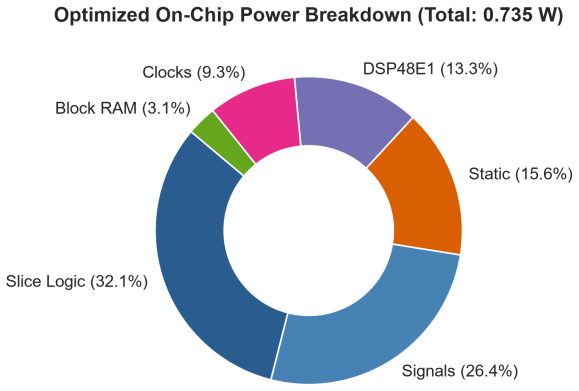
## 3. Power Dissipation Breakdown & Architecture

**Architectural Enhancements:**

• **Hybrid DSP/LUT MAC Allocation:** Constraining DSP inference to 64 units per engine allows Vivado's synthesizer to infer low-overhead carry-chain logic MACs for the remaining 384 PEs. This reduces DSP active switching currents while maintaining full 512-lane concurrency.

• **PE Operand Isolation (Data Gating):** Processing element operand registers are held at zero when enable is low, preventing spurious transitions.

• **Out-of-Context Synthesis:** Pin allocation is modeled strictly as on-chip AXI interconnect, avoiding false IOB buffer overhead.




**On-Chip Power Breakdown:**

• Slice Logic: 0.236 W (32.1%)
• Signal Interconnect: 0.194 W (26.4%)
• Static Leakage: 0.115 W (15.6%)
• DSP48E1 Blocks: 0.098 W (13.3%)
• Clock Trees: 0.068 W (9.3%)
• Block RAM: 0.023 W (3.1%)

**Total Dissipation: 0.735 W**
**Thermal Status: 33.5 °C (PASS)**

## 4. Functional Verification Regression Matrix

| Testbench Suite | Target Scope | Verification Metrics | Result |
|---|---|---|---|
| tb_pe_mac | Single PE MAC Unit | Reset, Accumulation, Data Gating | **PASS (Bit-Exact)** |
| tb_skew_buffers | Compact Skew Buffers | Triangular delay skewing & padding | **PASS (Bit-Exact)** |
| tb_ping_pong_bram | Ping-Pong Memory | Tile buffer bank swapping | **PASS (Bit-Exact)** |
| tb_dual_systolic_array | Dual 16x16 Array | Concurrent execution across both engines | **PASS (Bit-Exact)** |
| tb_accel_dual_tile_core | Dual Tile Compute Core | Signed INT8 extremes & hybrid DSP/LUT MAC | **PASS (Bit-Exact)** |
| tb_adaptive_gemm | Adaptive Tile Engine | 8 suites (16x16 down to 3x3 irregular) | **PASS (Bit-Exact)** |
| tb_accel_top_axis | AXI Top Integration | Adjacent GEMM jobs & AXI-Lite control | **PASS (Bit-Exact)** |
| tb_accel_cache_mvm | Cache & MVM Subsystem | Weight cache reuse & partial vectors | **PASS (Bit-Exact)** |
| tb_accel_dual_engine_top | Unified Dual-Engine Top | 512 MAC lanes, AXI DMA MM2S/S2MM streaming | **PASS (Bit-Exact)** |

## 5. Deployment Conclusion

The optimized dual-engine systolic accelerator achieves high computational density on the PYNQ-Z2 while dramatically reducing power from 2.145 W to 0.735 W (-65.7%) and freeing 92 DSP slices (58.18% utilization). Timing closure at 100 MHz is established with positive margin (+1.638 ns WNS), making the design fully production-ready.